

- during training
 - each neuron is dropped with probability p
 - rescale the remaining activations
 - e.g., activations scaled by $\frac{1}{1-p}$
 - effectively train an ensemble of subnetworks
 - during inference
 - dropout: turned off → deterministic predictions
 - no scaling is needed
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Linear Algebra

Basics

Linearity

- the operation preserving addition and scalar multiplication
- allows models to be analyzed using linear algebra
 - makes optimization more tractable
 - linear models → convex objectives
- nonlinearity can be added via activation functions

Basis

- a set of linearly independent vectors spanning a vector space
 - defines a coordinate system for representing vectors
 - special bases: eigenvectors, singular vectors
 - eigen decomposition $\approx$ change of basis

Rank

- dimension of the space spanned by the columns (or rows) of a matrix
 - intrinsic dimensionality of a transformation
- number of linearly independent columns (or rows)

Hessian

- matrix of second-order partial derivatives
- can capture curvature information of the loss surface
- **Symmetric matrix**
 - equal to its transpose
 - all eigenvalues $\rightarrow$ real number
 - all eigenvectors $\rightarrow$ orthogonal basis

Pseudo-Inverse

- applicable to any non-square matrix
 - standard inverse matrix $\rightarrow$ full rank & square matrix
- defined via SVD (A^+)
 - $A = U\Sigma V^T$
 - $$A^+ = V\Sigma^+U^T$$
 - Σ : invert only singular value
- gives a solution to a minimum-square problem

- $\min_x \|Ax - b\|^2 \Rightarrow x = A^+b$

Determinant

- a scalar value associated with a square matrix
- measures volume scaling of a linear transformation
- determinant = 0 $\leftrightarrow$ rank-deficient $\leftrightarrow$ non-invertible
 - from eigen decomposition $\rightarrow$ determinant = product of eigenvalues

Taylor Expansion

- local approximation of a function using its derivatives
 - around a point using polynomial terms
- First-order (linear approximation)
 - $f(x) \approx f(x_0) + \nabla f(x_0)^\top (x - x_0)$
 - giving the direction of steepest local change
- Second-order (quadratic approximation)
 - capture curvature by using Hessian
 - **Newton method**
 - $w_{t+1} = w_t - H^{-1} \nabla L(w_t)$
 - Hessian matrix H of loss function $\rightarrow$ eigenvalue $\approx$ steepness
 - able to control the step size
 - steep $\uparrow \rightarrow$ size $\downarrow$
 - **v.s. Gradient descent**
 - gradient descent: single learning rate for all directions
 - advantage: zig-zag $\downarrow$ & quadratic convergence $\uparrow$
 - drawback: computation of second-order matrix $\uparrow$

Norm

- measure the magnitude or length of a vector
- **L2 norm**
 - $\|w\|_2^2 = \sum_i w_i^2$
 - Euclidean distance
 - smooth → differentiable
 - discourage large weights → stable optimization
 - make magnitudes of parameters smaller
 - may be sensitive to outliers
 - Gaussian prior (MAP perspective)
- **L1 norm**
 - $\|w\|_1 = \sum_i |w_i|$
 - promote sparsity
 - useful where there is fewer features
 - implicit feature selection → robust to outliers but unstable optimization
 - non-differentiable at 0
 - Laplace prior (MAP perspective)

Cosine Similarity

$$\cos(x, y) = \frac{x^\top y}{\|x\| \|y\|}$$

- measure how aligned or related two vectors are
 - focus on directions (rather than magnitude)
- scale-invariant: invariant to absolute scale
- valid range from -1 to 1
- **v.s. dot product**
 - dot product is affected by magnitude
- cons
 - unstable if vector norm is very small

- less discriminative in dense representations
 - many vectors may have similar angles

Manifold

- space that is locally Euclidean → approximated by a tangent plane
 - tangent vectors: derivatives of curves
 - like a linear subspace in a sufficiently small neighborhood
- real-world data → on a much lower-dimensional manifold within a high-dimensional space
 - explaining why dimensionality reduction & representation learning works

Kernel

- function that computes inner products in an implicit feature space
 - without explicitly mapping data into high-dimensional feature space
 - implications
 - many problems become linearly separable in a higher-dimensional space
 - allow measuring similarity in a high-dimensional space efficiently
 - without paying the computational cost of explicit feature expansion
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Bayes Theorem

$$p(\theta \mid x) = \frac{p(x \mid \theta) p(\theta)}{p(x)}$$

- reveal how to update our belief about model parameters θ after observing data x
 - $p(x \mid \theta)$: likelihood
 - $p(\theta)$: prior

Matrix Decomposition

- decomposing a matrix into simpler components
- reveals principal directions & effective dimensionality
- useful for understanding optimization geometry & parameter efficiency

Diagonalization

$$A = PDP^{-1}$$

- transforming a matrix into a diagonal form via a similarity transform
- diagonal matrix D
 - contains scaling factors along independent directions
- change of coordinate system
 - where the linear transformation becomes independent per dimension
- diagonalizable: the matrix has a full set of linearly independent eigenvectors
- allow per-direction analysis of curvature and step size

Eigen Decomposition

diagonalizable

- $A = Q\Lambda Q^{-1}$
 - applicable only to square matrices
- similar to the change of basis
 - transforms the problem into independent directions
- **Eigenvectors**: principal directions of transformation
- **Eigenvalues**: scaling along each direction
 - Hessian analysis: eigenvalues $\approx$ curvature
 - second-order curvature from Taylor expansion
 - along principal directions of the loss surface

$$\det(A - \lambda I) = 0 \rightarrow \lambda = ?$$

$$(A - \lambda I)v = 0 \rightarrow v = ?$$

- large eigenvalue → steep direction
- **Spectral Theorem**
 - all of the real symmetric matrix can be decomposed by eigen decomposition
 - where leftmost and rightmost matrices have orthonormal eigenvector
 - Hessian eigen decomposition

Singular Value Decomposition (SVD)

- $W = U\Sigma V^T$
- sum of rank-1 component matrices, each weighted by a singular value
 - applicable to any matrix
 - singular vectors U, V : having orthonormal columns
- **Singular values**: importance (energy) of each rank-1 component
- **Singular vectors**: dominant input & output directions
- **Truncated SVD**
 - keep top- r singular values and corresponding singular vectors
 - discard small singular values → often corresponding to noise
 - suppress directions associated with small singular values
 - optimal low-rank approximation that minimizes mean squared reconstruction error
- widely used for compression and dimensionality reduction
 - e.g., PCA (Principal Components Analysis)

Low-Rank Adaptation (LoRA)

- assumption: weight updates lie in a low-rank subspace
- instead of updating full weight matrix
 - freezing pretrained weight matrix W
 - updating $\Delta W = BA$ with a small rank

- provides convergence guarantees for first-order methods
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Principal Component Analysis (PCA)

- a linear dimensionality reduction technique
- addresses the observation that
 - high-dimensional data often lies near a low-dimensional subspace
- **Goals**
 - represent data with fewer dimensions
 - while preserving as much information as possible
- PCA finds a low-dimensional linear subspace that maximizes variance
- **Data preprocessing**
 - $X \leftarrow X - \mu$
 - **centering** the data
 - assuming zero-mean data
- **Covariance Matrix** *+ Standardization*
 - $\Sigma = \frac{1}{N} X^\top X$
 - measures correlations between features
 - $\text{Var}(Xw) = w^\top \Sigma w$
 - variance along a direction w
- **Optimization**
 - first principal component solves:
 - $\max_{\|w\|=1} w^\top \Sigma w$ *covariance matrix에 대해 eigen decomposition*
 - **maximize projected variance**
 - unit-norm constraint avoids trivial solutions
 - solution: *가장 높은 분산을 지닌 eigenvalue의 축으로 projection*
 - eigenvector of Σ with the largest eigenvalue